CREDIT RISK ANALYSIS AND RATING MODEL – PROJECT REPORT

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1. Introduction

This project presents a structured **Credit Risk Assessment and Rating Model** developed for ten companies in the Indian automobile sector. The objective is to evaluate the creditworthiness of each company using:

- Five years of historical financial data (FY 2019, FY 2022–2025)
- Key financial ratios
- A weighted scoring framework
- A custom internal credit rating scale
- Probability of Default (PD) estimation
- Expected Loss (EL) modelling
- Scenario-based stress testing

The analysis closely aligns with methods used in commercial banks, NBFCs, and rating agencies for evaluating corporate borrowers.

The data for this project was manually collected from **Tickertape**, which provided clean, structured financial statement values for all ten selected companies.

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2. Dataset Description

Five years of financial data were collected for each of the ten auto-sector companies suggested earlier.
For every company, **16 financial variables** were captured:

- Revenue
- COGS
- EBITDA
- EBIT
- PAT
- Total Assets
- Total Liabilities
- Total Debt
- Shareholder Equity
- Current Assets
- Current Liabilities

- Inventory
- Interest Expense
- Operating Cash Flow
- Capital Expenditure
- Free Cash Flow

With 10 companies × 5 years × 16 variables, over **800 data points** were manually entered into Excel.

1	Company	Year	Revenue	EBITDA	EBIT	PAT	Total Asset	otal Liabiliti	Total Debt	urrent Ass	urrent Liabiliti	Invetory	hareholder Equ	ntrest expeses	COGS	erating Cash Fl	pital Expenditu	Free Cash Flow
2	Maruti Suzuki	2019	88,786	13,720.50	10,699.70	7,649.10	63,968.70	16,859.00	8	12,372.70	14160.5	3322.6	47109.7	75.9	63494.9	6,600.90	4,872.40	1,728.50
3	Maruti Suzuki	2022	90236.4	7612.8	4823.8	3879.5	4655.5	19,322.00	32.90	16793.4	17,023.50	3,532.30	55333.5	126.6	70819.4	1840.5	3459.3	-1618.8
4	Maruti Suzuki	2023	119889.4	13335.8	10510.1	8211	84596.9	22,805.60	24.90	11,615.60	20107.3	4283.5	61791.3	187	92079.3	9251.4	6347.3	2904.1
5	Maruti Suzuki	2024	146205.8	22873.9	17618.1	13488.2	115304.1	29,668	68	22,633.80	25,952	5,318	85,636	193.6	107454.5	16801.1	9199.9	7601.2
6	Maruti Suzuki	2025	158179.1	25,422.40	19,814.20	14,500.20	1,31,971.20	35,731	59	29,524.20	30,571	6,913	96,240	194.2	116796.9	16,136.20	10,640.60	5495.6
7	Tata Motors	2019	3,05,280.69	-2,021.92	-25,612.55	-28,826.23	3,05,703.49	2,45,000.87	70,973.67	1,23,431.16	1,45,457.43	39,013.73	60,702.62	5,758.60	2,29,097.71	18,890.75	35,303.52	-16,412.77
8	Tata Motors	2022	281617.5	27144.14	2308.45	-11441.47	329061.49	280229.19	103721.61	146977.54	1,50,682.81	35,240.34	48,832.30	9311.86	212282.14	14282.83	15168.42	-885.59
9	Tata Motors	2023	352871.35	38479.77	13619.41	2414.29	334674.43	282074.92	96264.3	150700.69	155027.33	40755.39	52599.51	10225.48	267419.29	35,388.01	19,230.17	16,157.84
10	Tata Motors	2024	440393	62616	35377	31399	369521	276427	69819	168392	173616	47788	93094	7594	318500	67915	31414	36501
11	Tata Motors	2025	452193	66990	43734	27830	376973	254219	47859	160325	166684	47269	122754	5083	319160	63102	38042	25060
12	Mahindra & Mahind	2019	1,07,742.09	17,882.96	13,892.19	5,315.46	1,61,888.66	1,13,544.68	43,526.17	69,406.04	58,609.33	12,200.16	48,343.98	5,021.35	73,940.38	-4,347.29	7,340.01	-11,687.30
13	Mahindra & Mahind	2022	93545.23	17887.32	14379.82	6577.32	1,72,388.49	1,15,563.23	51,057.61	75,148.00	56,288.33	11,595.82	56,825.26	5018.05	61592.86	9,247.55	6,039.80	3,207.75
14	Mahindra & Mahind	2023	126971.91	24,246.74	19,889.93	10,281.50	2,04,283.13	1,37,201.02	57,909.89	91325.44	70,579.41	16,854.97	67,082.11	5,829.70	86,528.30	-7,074.02	6,304.57	-13378.59
15	Mahindra & Mahind	2024	142376.12	28189.78	23466	11268.64	234721.74	155190.67	70533.61	98265.64	75545.65	18590.47	79531.07	7488.21	96447.13	-5629.95	9945.99	-15575.94
16	Mahindra & Mahind	2025	162929.29	34236.66	28163.01	12929.1	276013.16	1,82,839.89	82,877.33	1,26,573.27	90,322.01	20,330.93	93,173.27	9,083.39	1,07,713.42	3,175.81	10,391.95	-7,216.14
17	Ashok Leyland	2019	33,795.77	5,049.46	4,373.90	2,078.70	39,125.60	29,304.96	13,009.65	16,501.47	15,329.85	3,077.42	9,820.64	1,502.24	25,475.60	-3,745.49	1,126.35	-4,871.84
18	Ashok Leyland	2022	26741.54	2535.42	1,669.46	-358.61	43,573.65	34,983.49	15,457.87	18,638.30	18,615.90	2540.55	8590.16	1,869.05	20,391.10	2,844.56	508.72	2,335.84
19	Ashok Leyland	2023	42499.33	5258.65	4358.43	1238.71	54,696.93	43,901.12	19,691.83	24,243.44	22,454.95	3,440.43	10,795.81	2,093.50	33,020.23	-4,499.26	929.10	-5,428.36
20	Ashok Leyland	2024	46766.7	8015.61	7088.32	2,483.52	67,595.11	55,780.52	26,846.92	29,126.54	26,679.93	4,008.01	11,814.59	2,982.25	33,815.55	-6,257.98	1,133.54	-7,391.52
21	Ashok Leyland	2025	49976.21	9613.19	8526.54	3106.8	81546.31	65701.45	36568.32	34240.24	26551.46	3986.08	15,844.86	3,930.21	33,975.47	128.47	1,648.08	-1,519.61
22	Eicher Motors	2019	10,498.85	3,604.90	3,304.62	2,202.73	11386.83	2,468.11	0.00	4,415.31	2,097.96	633.38	8,918.72	7.33	6,039.04	1,575.47	791.51	783.96
23	Eicher Motors	2022	10798.8	2673.22	2221.29	1676.6	16139.83	3531.83	35.32	5619.34	2893.37	1132.4	12,608.00	18.78	6,984.08	1,527.04	640.83	886.21
24	Eicher Motors	2023	15352.46	4353.86	3827.65	2913.94	19197.64	4,207.36	130.75	3,683.23	3,234.58	1278.44	14990.28	28.02	9355.94	2822.66	682.49	2140.17
25	Eicher Motors	2024	18059.37	5850.5	5252.9	4001.01	23115.13	5069.6	271.44	4151.33	3598.41	1409.64	18045.53	50.88	10,503.65	3,723.71	818.51	2,905.20
26	Eicher Motors	2025	20875.06	6716.74	5987.41	4734.44	27174.41	5,877.92	308.17	6,954.16	4,265.78	1,563.75	21,296.49	54.34	11,950.85	3,979.92	1,039.33	2,940.59
27	Bajaj Auto	2019	32,393.87	7,225.75	6,960.06	4,927.61	28834.41	5,600.59	124.52	7,073.16	4,873.78	961.51	23,233.82	4.48	23,226.00	2,486.86	184.31	2,302.55
28	Bajaj Auto	2022	35824.89	7930.1	7660.34	6165.87	35111.21	5251.55	122.77	10,365.61	4,688.50	1230.51	29859.66	8.66	25,612.66	4,197.21	550.75	3,646.46
29	Bajaj Auto	2023	38172.19	8167.34	7881.49	6060.21	35136.45	5774.9	124.23	9650.39	5270.74	1563.55	29361.55	39.51	27,828.65	5,277.42	1,110.20	4,167.22
30	Bajaj Auto	2024	46574.04	10465.17	10100.4	7708.24	39,343.69	10,381.27	759.17	11,798.37	9,091.23	1688.75	28962.42	60.36	33744.51	6,558.16	722.84	5,835.32
31	Bajaj Auto	2025	52468.96	11026.66	10612.43	7324.73	54,109.92	18,921.17	6,210.38	19,445.93	11,583.81	2077.36	35188.75	388.9	37,333.52	-1405.57	873.84	-2,279.41
32	Hero Motocorp	2019	34718.31	5765.92	5141.48	3,444.09	18,504.36	5,267.84	124.84	8,413.21	4,409.18	1,249.53	13,236.52	37.18	27,623.15	1,032.28	979.56	52.72
33	Hero Motocorp	2022	30106.29	3800.73	3111.21	2316.88	22,478.39	6,483.69	251.68	10,572.22	5,517.63	1472.41	15994.7	53.1	22997.19	2103.7	569.66	1534.04
34	Hero Motocorp	2023	34730.5	4665.89	3,968.50	2,809.96	23,917.03	7,136.40	239.76	9,435.33	6,012.62	1756.39	16780.63	104.88	26472.33	2613.84	604.25	2009.59
35	Hero Motocorp	2024	38643.16	5923.93	5166.57	3744.83	26152.79	8,318.22	199.13	10,315.45	7,048.37	1755.9	17834.57	76.37	28072.81	4923.07	787.89	4135.18
36	Hero Motocorp	2025	41967.5	6828.8	6004.21	4378.48	28,390.32	8,986.23	196.21	13,433.89	7,475.98	2013.34	19404.09	70.65	29966.18	4297.39	856.63	3440.76
37	TVS Motors	2019	20,187.13	2,187.96	1,746.25	704.67	16,696.49	13,280.55	4,909.16	8,331.67	8,184.79	1,291.57	3,415.94	663.4	15602.01	-917.94	815.98	-1,733.92
38	TVS Motors	2022	4390.67	2,749.87	2,007.01	756.82	26,919.10	21,866.09	7,708.82	12,887.63	13,906.30	1642.36	5053.01	940.22	18769.12	-1575.19	969.83	-2545.02
39	TVS Motors	2023	32111.99	4,162.78	3,303.92	1,328.67	35,024.54	29,114.69	9,627.59	17,007.23	19,218.70	1921.51	5909.85	1367.89	24007.93	-4404.83	1,340.77	-5,745.60
40	TVS Motors	2024	39250.56	5,605.76	4,630.64	1,686.37	42,024.00	34,512.89	13,133.00	21520.01	21,078.21	2248.4	7511.11	1,927.72	28,244.77	-1252.67	1,145.17	-2,397.84
41	TVS Motors	2025	44188.48	6,674.07	5,628.51	2,235.56	47,650.89	38,208.62	14,684.51	25,984.25	23,173.54	2416.58	9442.27	2093.25	30502.57	3502.92	2478.06	1024.86
42	MRF	2019	16,484.19	2,733.18	1,925.58	1,130.60	18441.2	7603.73	1,054.73	8,463.80	5464.43	2993.2	10837.47	273.4	12251.98	1,252.85	1,963.76	-710.91
43	MRF	2022	19,633.71	2,375.78	1,170.73	669.23	23,059.74	9,027.84	1,168.08	10,263.65	6956.73	4129.67	14031.9	262.80	15,870.78	-578.01	1,707.01	-2,285.02
44	MRF	2023	23,261.17	2,649.21	1,396.17	768.93	24,369.41	9,661.59	1,332.20	9,241.17	7495.32	4141.05	14707.82	326.45	18,517.92	2,755.47	3,291.24	-535.77
45	MRF	2024	25,486.05	4,578.42	3,148.45	2081.22	26,849.43	10,146.27	1,379.30	10,549.55	7710.03	4468.58	16703.16	361.03	18212.16	3303.36	2163.5	1139.86
46	MRF	2025	28561.29	4500.32	2846.77	1869.28	29566.99	11,078.35	1,129.74	13,396.40	8959.57	5624.78	18488.64	367.77	21,617.16	1,867.80	1,308.17	559.63
47	Apollo Tyres	2019	17672.08	1899.42	1086.75	679.83	20150.9	10111.08	4166.33	5888.6	4374.42	3484.09	10039.82	198.63	13,527.09	1,071.12	2,293.17	-1,222.05
48	Apollo Tyres	2022	21071.16	2712.52	1312.85	638.61	27041.3	15289.17	5048.93	8310.97	8104.8	4155.39	11752.13	465.16	16,368.28	2153.56	1,846.24	307.32
49	Apollo Tyres	2023	24632.04	3396.95	1977.81	1,045.85	27281.11	14702.91	4404.02	8847.65	8004.19	4428.46	12578.2	550.62	18353.18	2,136.67	776.85	1,359.82
50	Apollo Tyres	2024	25531.65	4545.06	3067.23	1721.86	26874.9	12972.71	3386.58	9027.72	7043.82	4245.73	13902.19	527.03	17,188.08	3,439.52	730.42	2,709.10
51	Apollo Tyres	2025	26222.57	3515.48	2017.11	1121.33	27226.7	12461	2616.83	9815.34	7360.3	5131.16	14765.7	470.45	19,285.70	1,823.14	768.61	1,054.53

3. Ratio Framework

From the raw data, the following credit-relevant ratios were computed:

A. Leverage Ratios

- **Debt-to-Equity = Total Debt / Equity**
- **Debt-to-Assets = Total Debt / Total Assets**

B. Liquidity Ratios

- **Current Ratio = Current Assets / Current Liabilities**
- **Quick Ratio = (Current Assets – Inventory) / Current Liabilities**

C. Profitability Ratios

- **ROE = PAT / Equity**
- **ROA = PAT / Total Assets**
- **EBITDA Margin = EBITDA / Revenue**

- **Net Profit Margin = PAT / Revenue**

D. Efficiency Ratios

- **Asset Turnover = Revenue / Total Assets**
- **Inventory Turnover = COGS / Inventory**

E. Coverage Ratio

- **Interest Coverage = EBIT / Interest Expense**

This ratio framework covers all five fundamental areas of credit analysis used by professionals.

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4. Ratio Scoring Methodology

Because individual years may fluctuate, each ratio was converted to a **score between 1 and 5** using:

Weighted Ratio Score = 70% (Latest Year) + 30% (5-Year Average)

This approach stabilizes the score while giving more importance to recent performance.

Each ratio was scored between **1 (weak)** and **5 (excellent)** based on relative positioning across companies.

Company/Ratios	Yea	let Profit Margi	ROE	ROA	Jebt to Equit	terest Coverag	Debt to Asse	Current Rati	Quick Rati	sset Turnove	ventory Turnove	CF to Debt	evenue CAG
Maruti Suzuki	###	0.08615229	0.16	0.1196	0.00016982	180.770751	0.000125061	0.8737474	0.639109	1.387956923	19.11000421	825.1125	
Maruti Suzuki	###	0.042992628	0.07	0.8333	0.00059458	60.13270142	0.00706691	0.9864834	0.778988	19.38275158	20.04908983	55.94224924	
Maruti Suzuki	###	0.068488123	0.13	0.0971	0.00040297	71.3144385	0.000294337	0.5776807	0.364649	1.417184318	23.56234806	371.5421687	
Maruti Suzuki	###	0.09225489	0.16	0.117	0.00079056	118.1503099	0.000587143	0.8721476	0.667225	1.268001745	22.38262373	248.1698671	
Maruti Suzuki	###	0.091669506	0.15	0.1099	0.0006089	130.9083419	0.000444036	0.9657742	0.739635	1.198588025	19.09803537	275.3617747	
Average + Trend		0.087062101	0.15	0.1535	0.00058024	125.3124319	0.000821875	0.9325919	0.709121	2.318280573	19.62075083	299.3209559	0.20574586
Tata Motors	###	-0.09442533	-0.5	-0.094	1.16920275	-0.351113118	0.232165063	0.8485724	0.580358	0.998616961	5.872232929	0.266165608	
Tata Motors	###	-0.040627695	-0.2	-0.035	2.12403696	2.915007313	0.315204341	0.9754101	0.741539	0.855820291	6.023839157	0.137703512	
Tata Motors	###	0.006841842	0.05	0.0072	1.83013682	3.763126034	0.287635658	0.9720911	0.709199	1.054372006	7.037745147	0.36761302	
Tata Motors	###	0.071297682	0.34	0.085	0.74998389	8.24545694	0.188944607	0.9699106	0.694659	1.191794242	7.194212916	0.972729486	
Tata Motors	###	0.061544517	0.23	0.0738	0.38987732	13.17922487	0.126956042	0.96185	0.678265	1.199536837	6.715128817	1.318498088	
Average + Trend		0.043359023	0.15	0.0539	0.64870838	10.89055953	0.157923572	0.956965	0.679027	1.157684206	6.67117971	1.106711245	0.17099417
Mahindra & Mahind	###	0.049335037	0.11	0.0328	0.90034312	3.561384887	0.268864848	1.1842149	0.976054	0.665532039	6.060607402	-0.099877614	
Mahindra & Mahind	###	0.070311656	0.12	0.0382	0.898502	3.564595809	0.296177604	1.3350547	1.129047	0.542641971	5.311643334	0.181119915	
Mahindra & Mahind	###	0.080974603	0.15	0.0503	0.86326876	4.159174572	0.283478572	1.2939388	1.05513	0.621548681	6.082664137	-0.122155646	
Mahindra & Mahind	###	0.079146981	0.14	0.048	0.88686862	3.764555214	0.300498837	1.3007452	1.054663	0.606574065	5.442004952	-0.079819394	
Mahindra & Mahind	###	0.079354056	0.14	0.0468	0.88949685	3.769150064	0.300265864	1.4013558	1.176262	0.590295369	5.534920121	0.038319405	
Average + Trend		0.077095179	0.14	0.0458	0.88895655	3.767536678	0.297143248	1.3718676	1.146853	0.594802285	5.580354481	0.021878784	0.14781331
Ashok Leyland	###	0.061507698	0.21	0.0531	1.32472527	3.361287145	0.332509917	1.0764274	0.87568	0.863776402	8.278233065	-0.287900904	
Ashok Leyland	###	-0.013410222	-0	-0.008	1.79948569	1.356528718	0.3547527	1.0012033	0.864731	0.613708973	8.026254158	0.184020179	
Ashok Leyland	###	0.029146577	0.11	0.0226	1.82402525	2.511893957	0.360017098	1.0796479	0.926433	0.776996625	11.04174567	-0.228483589	
Ashok Leyland	###	0.053104453	0.21	0.0367	2.27235308	2.687772655	0.397172517	1.0917023	0.941477	0.691865136	9.079901295	-0.23309862	
Ashok Leyland	###	0.062165578	0.2	0.0381	2.30789796	2.445973625	0.448436232	1.2895803	1.139454	0.612856793	8.500146984	0.0035131	
Average + Trend		0.05506675	0.18	0.0352	2.1872378	2.453988903	0.42747867	1.2350199	1.082484	0.642551991	8.645679759	-0.031257782	0.13928835
Eicher Motors	###	0.209806788	0.25	0.1934	0	491.8008186	0	2.104573	1.80267	0.922016927	9.534623765	0	
Eicher Motors	###	0.155257992	0.13	0.1039	0.0028014	142.343983	0.002188375	1.9421436	1.550766	0.669077679	6.167502649	43.23442809	
Eicher Motors	###	0.189802807	0.19	0.1518	0.00872232	155.3840114	0.006810733	1.1387043	0.743463	0.799705589	6.961057707	21.5882218	
Eicher Motors	###	0.221547596	0.22	0.1731	0.01504195	114.9862421	0.011742958	1.1536568	0.761917	0.781279188	7.065100777	13.71835396	
Eicher Motors	###	0.226798869	0.22	0.1742	0.01447046	123.6058152	0.011340449	1.63022	1.26364	0.768188159	9.465045183	12.91468994	
Average + Trend		0.218952051	0.22	0.1697	0.01259149	148.2113229	0.009863265	1.6193119	1.251895	0.774147764	8.977131433	14.52762459	0.25746334
Bajaj Auto	###	0.152115508	0.21	0.1709	0.00535943	1612.890625	0.004318451	1.4512678	1.253986	1.12344487	24.15575501	19.97157083	
Bajaj Auto	###	0.172111345	0.21	0.1756	0.00411157	915.7159353	0.003496604	2.2108585	1.948406	1.020326272	20.81467034	34.18758654	
Bajaj Auto	###	0.15875982	0.21	0.1725	0.00423104	206.7157682	0.003535645	1.8309365	1.534289	1.086398597	19.91986572	42.48104323	
Bajaj Auto	###	0.165505075	0.27	0.1959	0.02621224	173.3792247	0.019295852	1.2977749	1.112019	1.183774069	20.75116687	8.638592147	
Bajaj Auto	###	0.139601204	0.21	0.1354	0.17648766	28.35345847	0.114773409	1.6787162	1.499383	0.969673583	19.82603801	-0.226325925	
Average + Trend		0.14500642	0.21	0.1458	0.13652548	196.0707216	0.089066584	1.6832746	1.490453	1.001788552	20.20627636	6.144719862	0.17439233
Hero Motocorp	###	0.099200969	0.26	0.1861	0.00943148	155.0812265	0.006746518	1.9081122	1.624719	1.876223225	22.10683217	8.268824095	
Hero Motocorp	###	0.076956676	0.14	0.1031	0.01573521	71.57683616	0.011196531	1.9160799	1.649224	1.339343699	15.6187407	8.358630006	
Hero Motocorp	###	0.08090756	0.17	0.1175	0.0142879	44.48789092	0.010024656	1.5692543	1.277137	1.452124281	16.3976276	10.9019019	
Hero Motocorp	###	0.096907965	0.21	0.1432	0.01116539	77.56880974	0.007614102	1.4635228	1.214401	1.477592257	15.98547386	24.72289459	
Hero Motocorp	###	0.104330256	0.23	0.1542	0.01011179	96.65675867	0.006911158	1.7969403	1.527632	1.478232722	15.9003831	21.90199276	
Average + Trend		0.100529384	0.22	0.1502	0.01072216	94.38202239	0.007387389	1.7770928	1.50693	1.492173876	16.29081162	19.78064954	0.06524991
TVS Motors	###	0.034906894	0.21	0.0422	1.43713297	3.298100693	0.294023474	1.0179455	0.860144	1.2090643	12.07987953	-0.186985146	
TVS Motors	###	0.172370048	0.15	0.0281	1.5255897	2.924709111	0.286369901	0.9267476	0.808646	0.163106122	11.42813999	-0.204336072	
TVS Motors	###	0.041376134	0.22	0.0379	1.62907519	3.043212539	0.274881269	0.8849313	0.78495	0.916842591	13.47295496	-0.45752156	
TVS Motors	###	0.042964228	0.22	0.0401	1.74847659	2.907974187	0.312511898	1.02096	0.914291	0.934003427	13.54694466	-0.095383385	
TVS Motors	###	0.050591466	0.24	0.0469	1.55518853	3.188376926	0.308168641	1.1212896	1.017008	0.927337978	13.07725649	0.238545243	
Average + Trend		0.055946553	0.23	0.0446	1.56235975	3.153606255	0.304275359	1.0832152	0.975008	0.89815785	12.97039008	0.124640815	0.29840893
MRF	###	0.068586931	0.1	0.0613	0.09732253	9.997000732	0.057194217	1.5488898	1.001129	0.893878381	4.093271415	1.187839542	
MRF	###	0.034085764	0.05	0.029	0.08324461	9.040258752	0.050654517	1.4753555	0.881733	0.851428073	3.843110951	-0.494837682	
MRF	###	0.033056377	0.05	0.0316	0.09057767	8.115209067	0.054666896	1.2329253	0.680441	0.954523314	4.477946297	2.068360607	
MRF	###	0.08166114	0.12	0.0775	0.08257719	12.68155001	0.051371668	1.3682891	0.788709	0.949221268	4.230648704	2.394953962	
MRF	###	0.065448024	0.1	0.0632	0.06110455	12.23677842	0.038209503	1.4952057	0.86741	0.965985716	4.283441787	1.653300759	
Average + Trend		0.062783911	0.1	0.06	0.06766278	11.68999271	0.04187246	1.4738839	0.860352	0.953092206	4.254114401	1.565887563	0.20107484
Apollo Tyres	###	0.038469156	0.07	0.0337	0.41498055	9.562603836	0.206756522	1.3461442	0.549675	0.876987132	3.882531737	0.257089573	
Apollo Tyres	###	0.030307302	0.05	0.0236	0.42961829	5.831369851	0.186711807	1.025438	0.512731	0.779221413	3.939047839	0.4265379	
Apollo Tyres	###	0.042458927	0.08	0.0383	0.35013118	6.169318223	0.161431115	1.1053773	0.55211	0.902897279	4.276211723	0.485163555	
Apollo Tyres	###	0.067440216	0.12	0.0641	0.24360047	8.623911352	0.126012748	1.2816511	0.678892	0.950018419	3.963039777	1.01563229	
Apollo Tyres	###	0.042762018	0.08	0.0412	0.17722357	7.472590073	0.096112639	1.3335516	0.636412	0.963119658	4.113453394	0.69669713	
Average + Trend		0.043219669	0.08	0.0409	0.22098974	7.490400652	0.113900338	1.2990159	0.621277	0.942518395	4.089874444	0.660555813	0.14058904

5. Category Scores

Ratios were grouped into four major risk categories:

Category	**Ratios Included**
Leverage	D/E, D/A
Profitability	ROE, ROA, EBITDA Margin, Net Margin
Liquidity	Current Ratio, Quick Ratio
Efficiency	Asset Turnover, Inventory Turnover

Each category score = **average of its ratio scores**.

This produced four well-balanced category-level credit scores for each company.

Company/Ratios	Net Profit Margin	ROE	ROA	Debt to Equity	Interest Coverage	Debt to Asset	Current Ratio	Quick Ratio	Asset Turnover	Inventory Turnover	CF to Debt	Revenue CAGR
Maruti Suzuki	8.706210067	14.58791841	15.35200645	0.000580235	125.3124319	0.000821875	0.932591935	0.7091206	2.318280573	19.62075083	299.32096	20.57458611
Tata Motors	4.335902318	15.27426128	5.389421229	0.648708385	10.89055953	0.157923572	0.956965031	0.6790271	1.157684206	6.67117971	1.1067112	17.0994168
Mahindra & Mahindr	7.709517912	13.66998642	4.57597481	0.888956555	3.767536678	0.297143248	1.371867602	1.1468527	0.594802285	5.580354481	0.0218788	14.78133097
Ashok Leyland	5.506674986	17.87099721	3.521214382	2.187237804	2.453988903	0.42747867	1.235019884	1.0824841	0.642551991	8.645679759	-0.0312578	13.92883524
Eicher Motors	21.89520514	21.67200258	16.97425139	0.012591489	148.2113229	0.009863265	1.619311879	1.2518953	0.774147764	8.977131433	14.527625	25.74633438
Bajaj Auto	14.50064201	21.16658553	14.57733865	0.136525475	196.0707216	0.089066584	1.683274602	1.4904532	1.001788552	20.20627636	6.1447199	17.43923322
Hero Motocorp	10.05293844	21.84405786	15.02029157	0.010722157	94.38202239	0.007387389	1.777092805	1.5069296	1.492173876	16.29081162	19.78065	6.524991319
TVS Motors	5.594655266	22.82625288	4.45587006	1.56235975	3.153606255	0.304275359	1.083215183	0.9750077	0.89815785	12.97039008	0.1246408	29.84089319
MRF	6.278391134	9.657307296	6.001246264	0.067662776	11.68999271	0.04187246	1.473883908	0.8603523	0.953092206	4.254114401	1.5658876	20.10748388
Apollo Tyres	4.321966945	7.745895668	4.088606728	0.220989739	7.490400652	0.113900338	1.299015871	0.6212772	0.942518395	4.089874444	0.6605558	14.05890419

SCORES												
Company/Ratios	Net Profit Margin	ROE	ROA	Debt to Equity	Interest Coverage	Debt to Asset	Current Ratio	Quick Ratio	Asset Turnover	Inventory Turnover	evenue CAG	CF to Debt
Maruti Suzuki	2	3	4	5	5	5	2	2	5	4	5	5
Tata Motors	1	4	2	3	4	4	2	2	3	2	4	3
Mahindra & Mahindr	2	3	1	2	1	3	4	4	1	2	3	0
Ashok Leyland	2	4	1	1	1	3	4	4	1	2	3	0
Eicher Motors	5	5	4	5	5	5	5	5	1	2	3	5
Bajaj Auto	3	5	3	5	5	5	5	5	2	5	3	5
Hero Motocorp	3	5	4	5	5	5	5	5	3	4	2	5
TVS Motors	2	5	1	1	1	3	3	3	1	3	5	1
MRF	2	2	2	5	3	5	5	3	2	1	5	3
Apollo Tyres	1	2	1	4	2	4	4	2	2	1	3	2

Company/Ratios	Profitability Ratio	Leverage Score	Liquidity Score	Efficiency Score	Credit Score
Maruti Suzuki	3.00	5.00	2.00	4.67	3.90
Tata Motors	2.33	3.67	2.00	3.00	2.92
Mahindra & Mahindra	2.00	2.00	4.00	2.00	2.30
Ashok Leyland	2.33	1.67	4.00	2.00	2.27
Eicher Motors	4.67	5.00	5.00	2.00	4.45
Bajaj Auto	3.67	5.00	5.00	3.33	4.35
Hero Motocorp	4.00	5.00	5.00	3.00	4.40
TVS Motors	2.67	1.67	3.00	3.00	2.37
MRF	2.00	4.33	4.00	2.67	3.33
Apollo Tyres	1.33	3.33	3.00	2.00	2.48

6. Final Credit Score & Weighting System

A weighted credit score was calculated as follows:

- **Leverage – 40%**
- **Profitability – 30%**
- **Liquidity – 15%**
- **Efficiency – 15%**

Final Credit Score = 0.40×Leverage + 0.30×Profitability + 0.15×Liquidity + 0.15×Efficiency

This weighting method reflects real-world credit practices, where leverage and profitability drive the majority of rating outcomes.

7. Internal Credit Rating Scale

The final credit score was mapped to a rating using the following internal scale:

Score Range Rating

> 4.0 A+

3.5 – 4.0 A

3.0 – 3.49 B+

2.5 – 2.99 B

2.0 – 2.49 C+

< 2.0 C

This created a consistent and transparent internal rating for each company.

WEIGHTS	30%	40%	15%	15%	100%	
Company	**Profitability Ratio**	**Leverage Score**	**Liquidity Score**	**Efficiency Score**	**Credit Score**	**Credit Ratings**
Maruti Suzuki	3.00	5.00	2.00	4.67	3.90	A
Tata Motors	2.33	3.67	2.00	3.00	2.92	B
Mahindra & Mahindra	2.00	2.00	4.00	2.00	2.30	C+
Ashok Leyland	2.33	1.67	4.00	2.00	2.27	C+
Eicher Motors	4.67	5.00	5.00	2.00	4.45	A+
Bajaj Auto	3.67	5.00	5.00	3.33	4.35	A+
Hero Motocorp	4.00	5.00	5.00	3.00	4.40	A+
TVS Motors	2.67	1.67	3.00	3.00	2.37	C+
MRF	2.00	4.33	4.00	2.67	3.33	B+
Apollo Tyres	1.33	3.33	3.00	2.00	2.48	C+

Interval	Rating
>4.00	A+
3.5 - 4.00	A
3.0 - 3.49	B+
2.5 - 2.99	B
2.0 - 2.49	C+
1.5 - 1.99	C

8. Expected Loss (EL) Modeling

To quantify credit risk, the following assumptions were used:

- **Probability of Default (PD)** based on rating:
  - A+ = 0.5%
  - A = 1%
  - B+ = 2%
  - B = 3%
  - C+ = 5%
  - C = 8%
- **Loss Given Default (LGD) = 40%**
- **Exposure (EAD) = 100 (standardized)**

Expected Loss = PD × LGD × Exposure

This gives a comparable EL value for all companies, allowing ranking based on credit risk.

Based on Values of Credit Rating Companies	
Credit Ratings	PD
A+	0.001
A	0.002
B+	0.006
B	0.012
C+	0.04
C	0.1

Expected Loss (EL) = Probability of Default(PD) * Loss Given a Default * Exposure

Company	PD	LGD	EXPOSURE	EL9(Percentage)
Maruti Suzuki	0.002	40%	100	0.08
Tata Motors	0.012	40%	100	0.48
Mahindra & Mahindra	0.04	40%	100	1.6
Ashok Leyland	0.04	40%	100	1.6
Eicher Motors	0.001	40%	100	0.04
Bajaj Auto	0.001	40%	100	0.04
Hero Motocorp	0.001	40%	100	0.04
TVS Motors	0.04	40%	100	1.6
MRF	0.006	40%	100	0.24
Apollo Tyres	0.04	40%	100	1.6

Exposure	100	(Constant) Assumption
LGD	40%	Industry Average

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9. Scenario Analysis (Stress Testing)

Two stress scenarios were created, adjusting **category scores** rather than raw data — a clean and professional method used by banks.

Scenario 1: Mild Stress

- Leverage: –10%
- Profitability: –8%
- Liquidity: –5%
- Efficiency: –5%

Scenario 2: Severe Stress

- Leverage: –20%
- Profitability: –18%
- Liquidity: –10%
- Efficiency: –12%
- Coverage (if included): –25%

Under each scenario:

- Category scores were reduced
- Final weighted scores recalculated
- Rating downgrades observed

This demonstrated how sensitive each company's credit profile is under mild and severe economic shocks.

	Scenario 1	Scenario 2
Ratios	Mild Stress	Severe Stress
Profitability	-10%	-20%
Leverage	-5%	-10%
Liquidity	-8%	-18%
Efficiency	-5%	-12%

Interval	Rating
>4.00	A+
3.5 - 4.00	A
3.0 - 3.49	B+
2.5 - 2.99	B
2.0 - 2.49	C+
1.5 - 1.99	C

Scenario 1 Mild Stress

Company/Ratios	Profitability Ratio	Leverage Score	Liquidity Score	Efficiency Score	Credit Score	Credit Rating
Maruti Suzuki	2.70	4.75	1.84	4.43	3.65	B+
Tata Motors	2.10	3.48	1.84	2.85	2.73	C+
Mahindra & Mahindra	1.80	1.90	3.68	1.90	2.14	C
Ashok Leyland	2.10	1.58	3.68	1.90	2.10	C
Eicher Motors	4.20	4.75	4.60	1.90	4.14	A+
Bajaj Auto	3.30	4.75	4.60	3.17	4.06	A+
Hero Motocorp	3.60	4.75	4.60	2.85	4.10	A+
TVS Motors	2.40	1.58	2.76	2.85	2.19	C+
MRF	1.80	4.12	3.68	2.53	3.12	B
Apollo Tyres	1.20	3.17	2.76	1.90	2.33	C+

Scenario 2 Severe Stress

Company/Ratios	Profitability Ratio	Leverage Score	Liquidity Score	Efficiency Score	Credit Score	Credit Rating
Maruti Suzuki	2.40	4.50	1.64	4.11	3.38	B+
Tata Motors	1.87	3.30	1.64	2.64	2.52	B
Mahindra & Mahindra	1.60	1.80	3.28	1.76	1.96	C
Ashok Leyland	1.87	1.50	3.28	1.76	1.92	C
Eicher Motors	3.73	4.50	4.10	1.76	3.80	A
Bajaj Auto	2.93	4.50	4.10	2.93	3.74	A
Hero Motocorp	3.20	4.50	4.10	2.64	3.77	A
TVS Motors	2.13	1.50	2.46	2.64	2.01	C+
MRF	1.60	3.90	3.28	2.35	2.88	B
Apollo Tyres	1.07	3.00	2.46	1.76	2.15	C+

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10. Conclusion

This project successfully delivers a complete and professional-grade credit risk analysis framework, including:

- Collection of multi-year financial data
- Calculation of all essential credit ratios
- Weighted scoring mechanism
- A transparent internal rating model
- PD–LGD–EL based credit risk quantification
- Stress testing through scenario analysis